4-bit Array Multiplier

*~/array_multiplier_4bit.v - Mousepad

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```
`timescale 1ns/1ps
module array_multiplier_4bit (
    input  [3:0] A, // Multiplicand
    input  [3:0] B, // Multiplier
    output [7:0] P  // Product
);

    wire [3:0] pp0, pp1, pp2, pp3; // Partial products
    wire [7:0] sum1, sum2, sum3;

    // Generate partial products
    assign pp0 = A & {4{B[0]}};
    assign pp1 = A & {4{B[1]}};
    assign pp2 = A & {4{B[2]}};
    assign pp3 = A & {4{B[3]}};

    // Shift and add partial products
    assign sum1 = {4'b0000, pp0};
    assign sum2 = {3'b000, pp1, 1'b0};
    assign sum3 = {2'b00, pp2, 2'b00};
    wire [7:0] sum4 = {1'b0, pp3, 3'b000};

    // Final addition
    assign P = sum1 + sum2 + sum3 + sum4;

endmodule
```

*~/tb_array_multiplier_4bit.v - Mousepad

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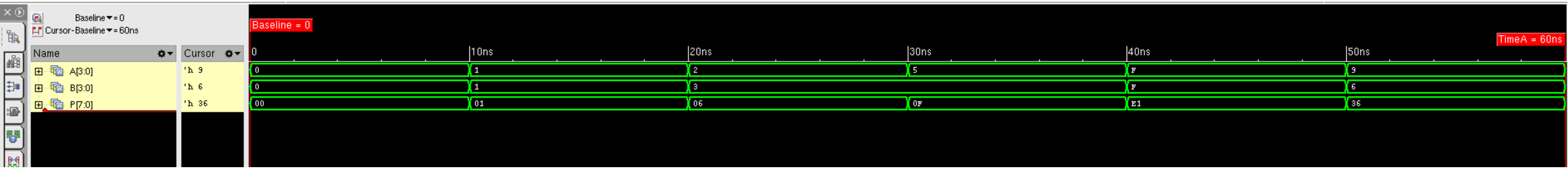
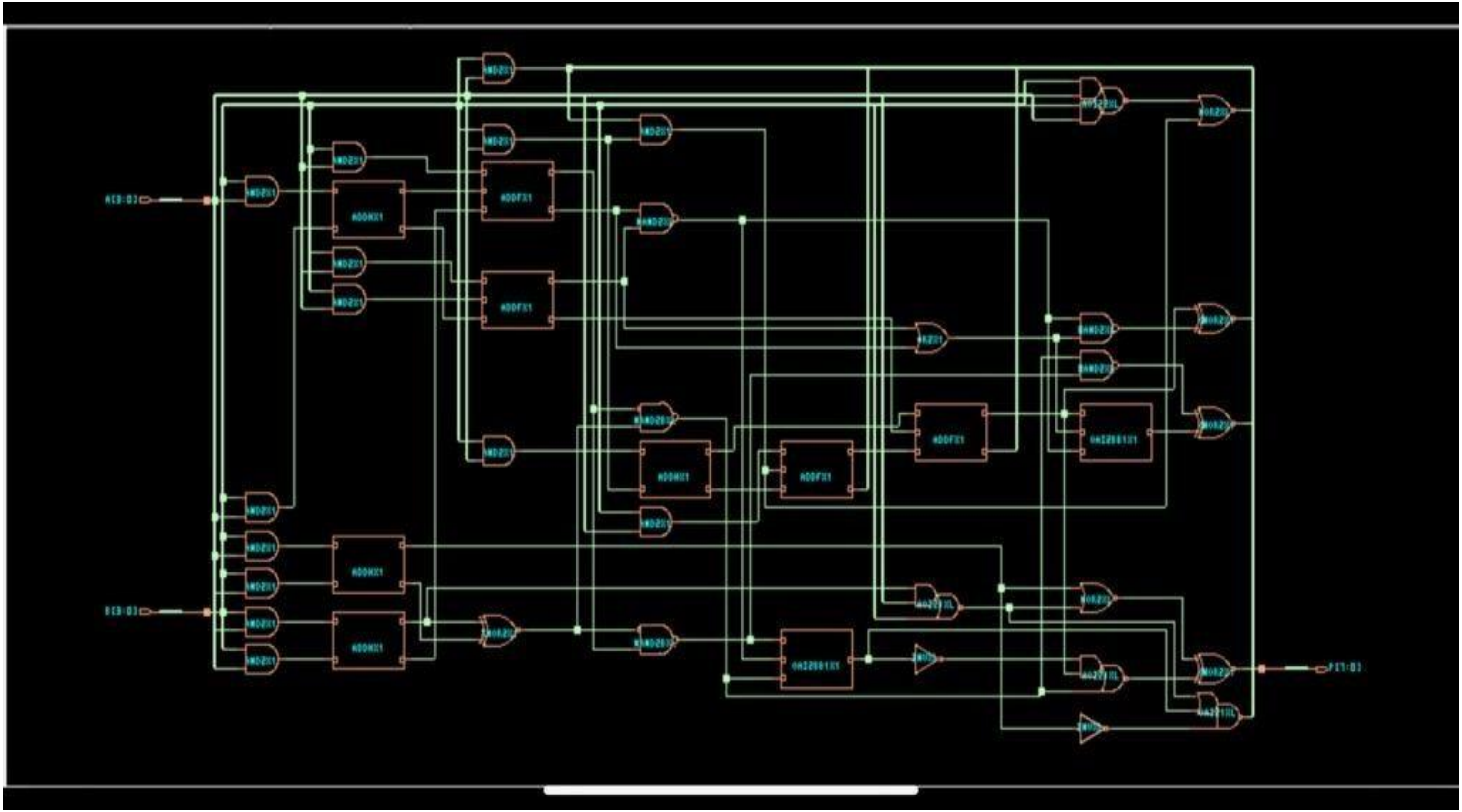
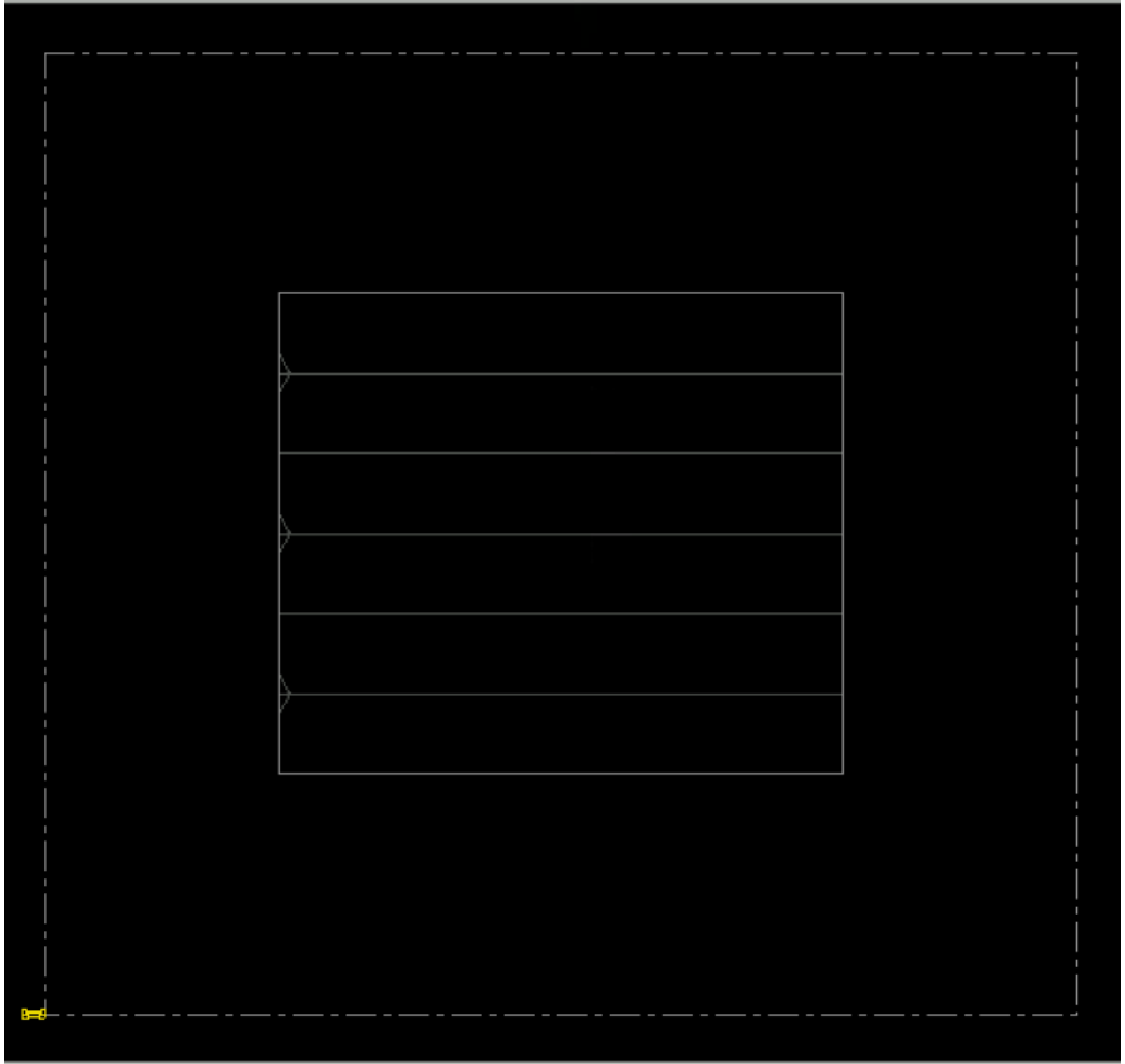
```
`timescale 1ns / 1ps

module tb_array_multiplier_4bit;

    // Inputs
    reg [3:0] A;
    reg [3:0] B;

    // Output
    wire [7:0] P;

    // Instantiate the Unit Under Test (UUT)
    array_multiplier_4bit uut (
        .A(A),
        .B(B),
        .P(P)
    );

    initial begin
        // Display header
        $display("Time\tA\tB\tProduct");
        $monitor("%g\t%b\t%b\t%b", $time, A, B, P);

        // Test cases
        A = 4'b0000; B = 4'b0000; #10;
        A = 4'b0001; B = 4'b0001; #10;
        A = 4'b0010; B = 4'b0011; #10;
        A = 4'b0101; B = 4'b0011; #10;
        A = 4'b1111; B = 4'b1111; #10;
        A = 4'b1001; B = 4'b0110; #10;

        // Finish simulation
        $finish;
    end

endmodule
```


Baseline = 0
Cursor-Baseline = 60ns
Name
Cursor
A[3:0]
B[3:0]
P[7:0]
'h 9
'h 6
'h 36
Baseline = 0
TimeA = 60ns
0
10ns
20ns
30ns
40ns
50ns
0 1 2 5 F 9
0 1 3 F 6
00 01 06 0F E1 36

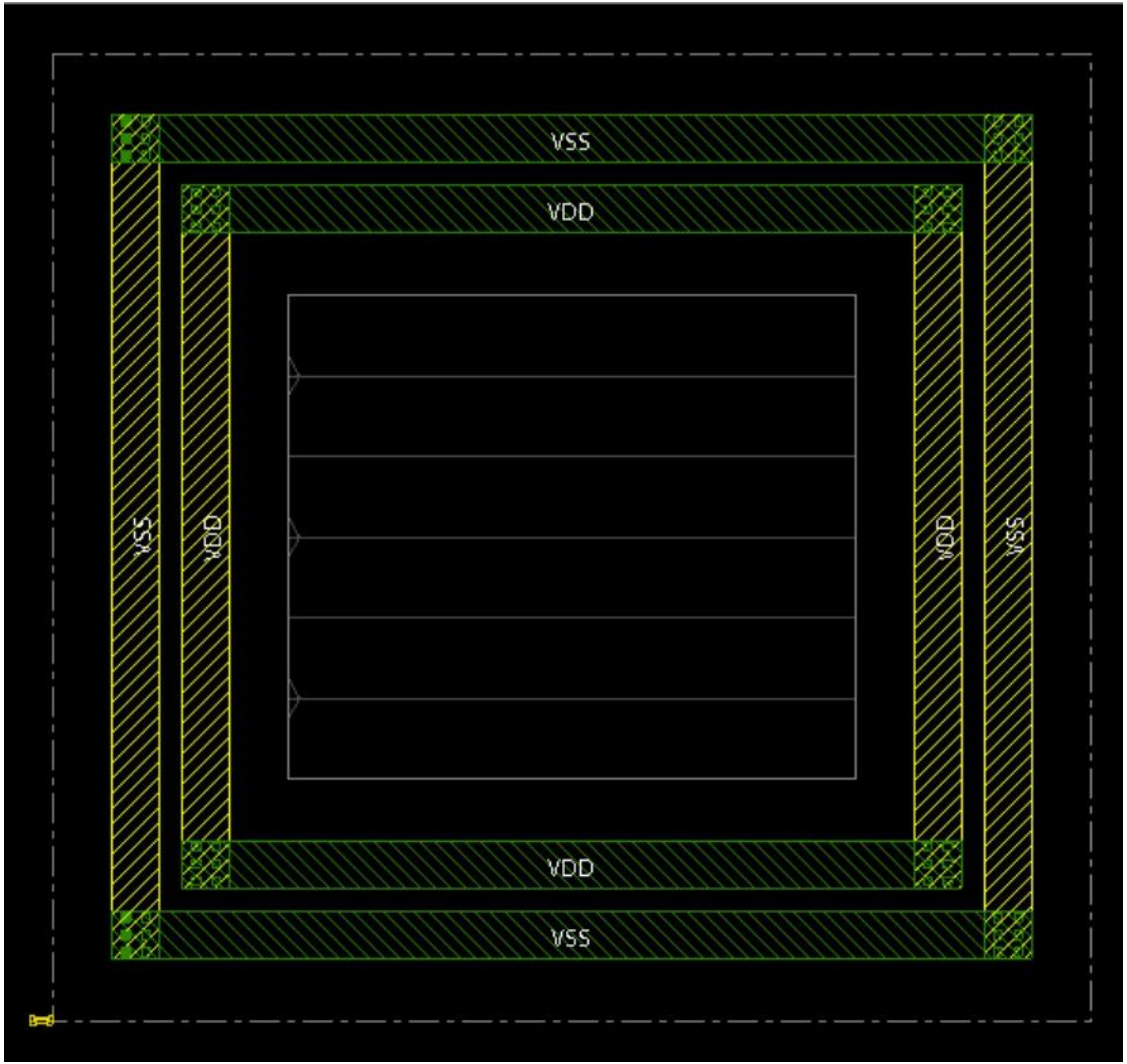
Schematic Diagram



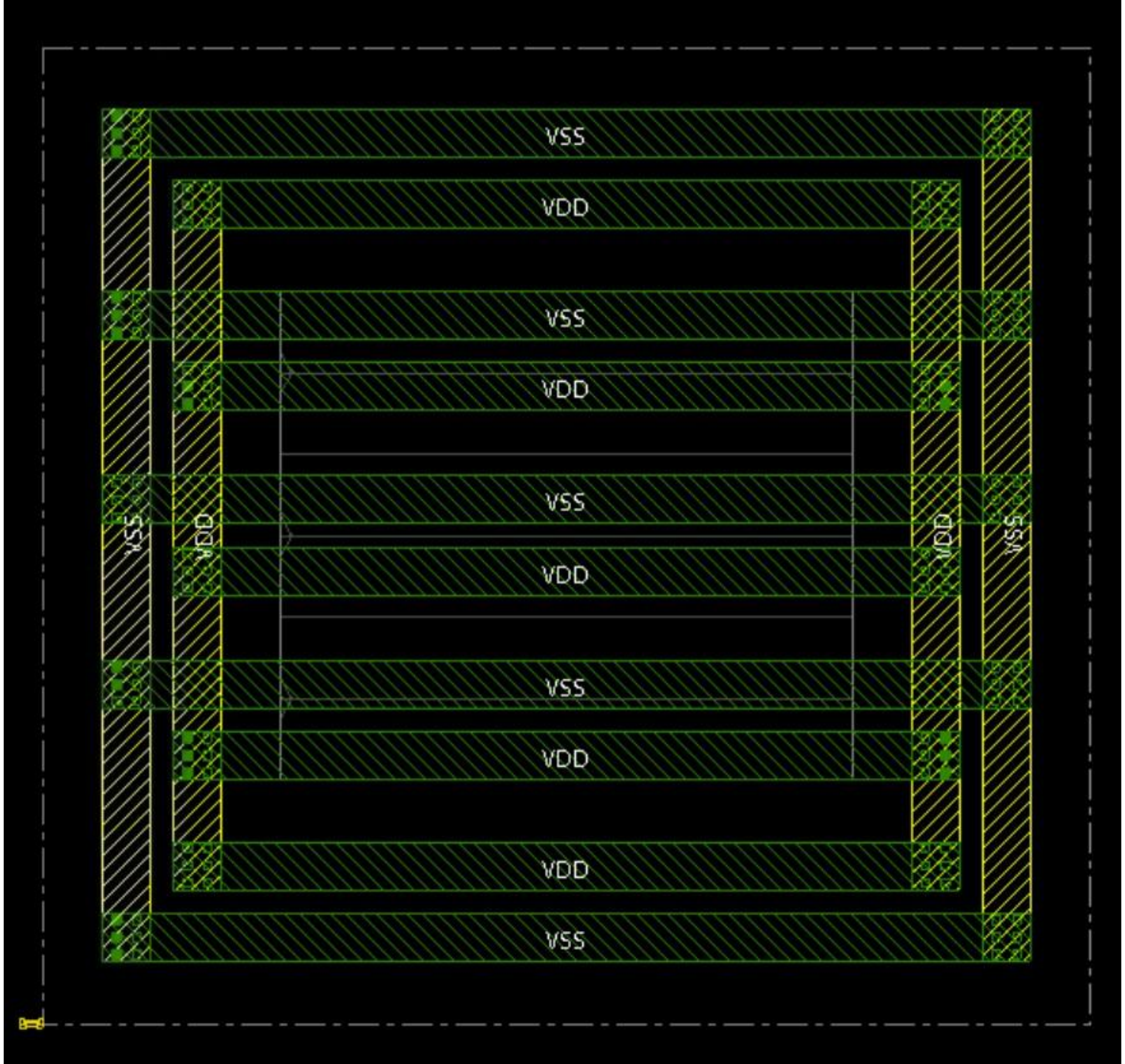
Floorplan :



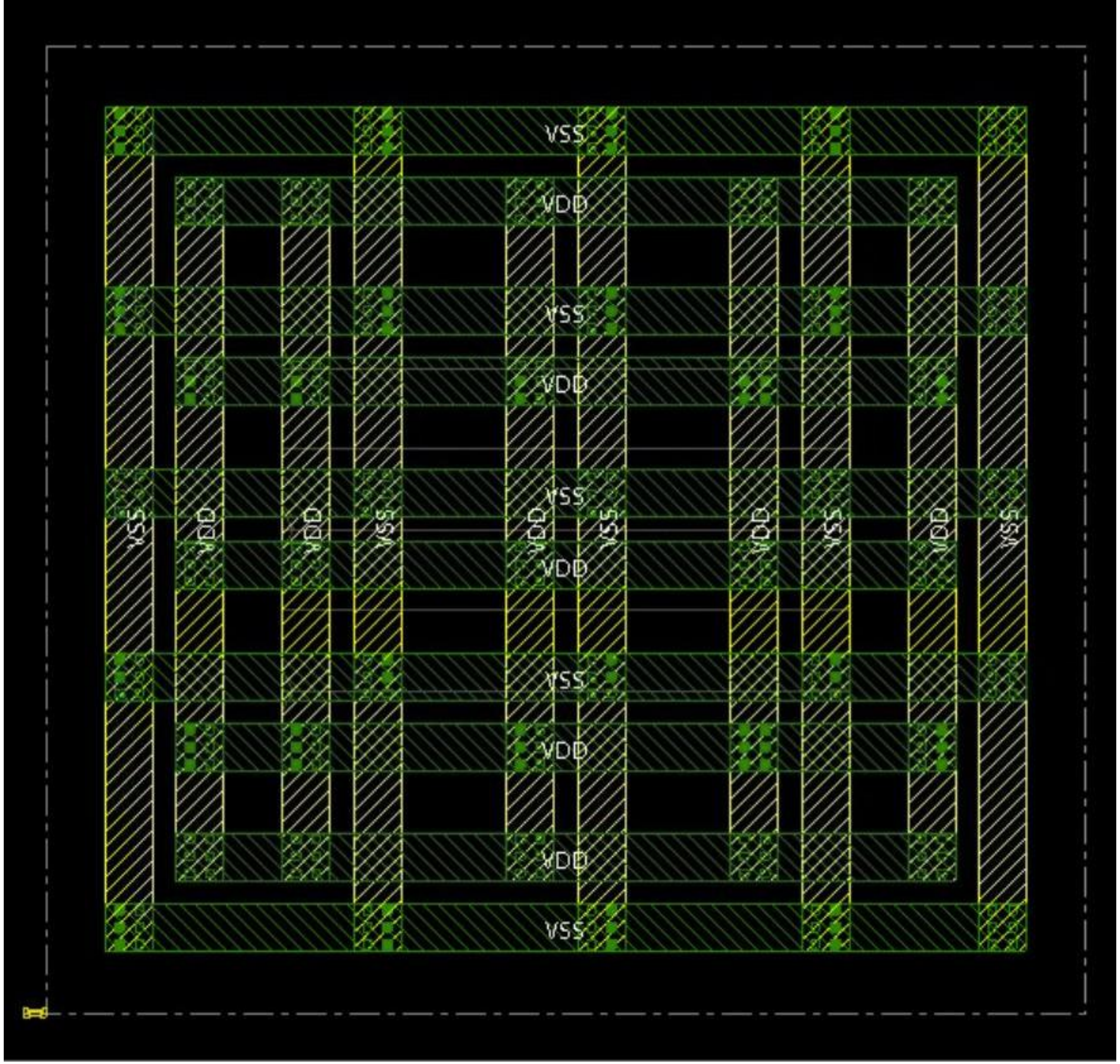
Power Rings



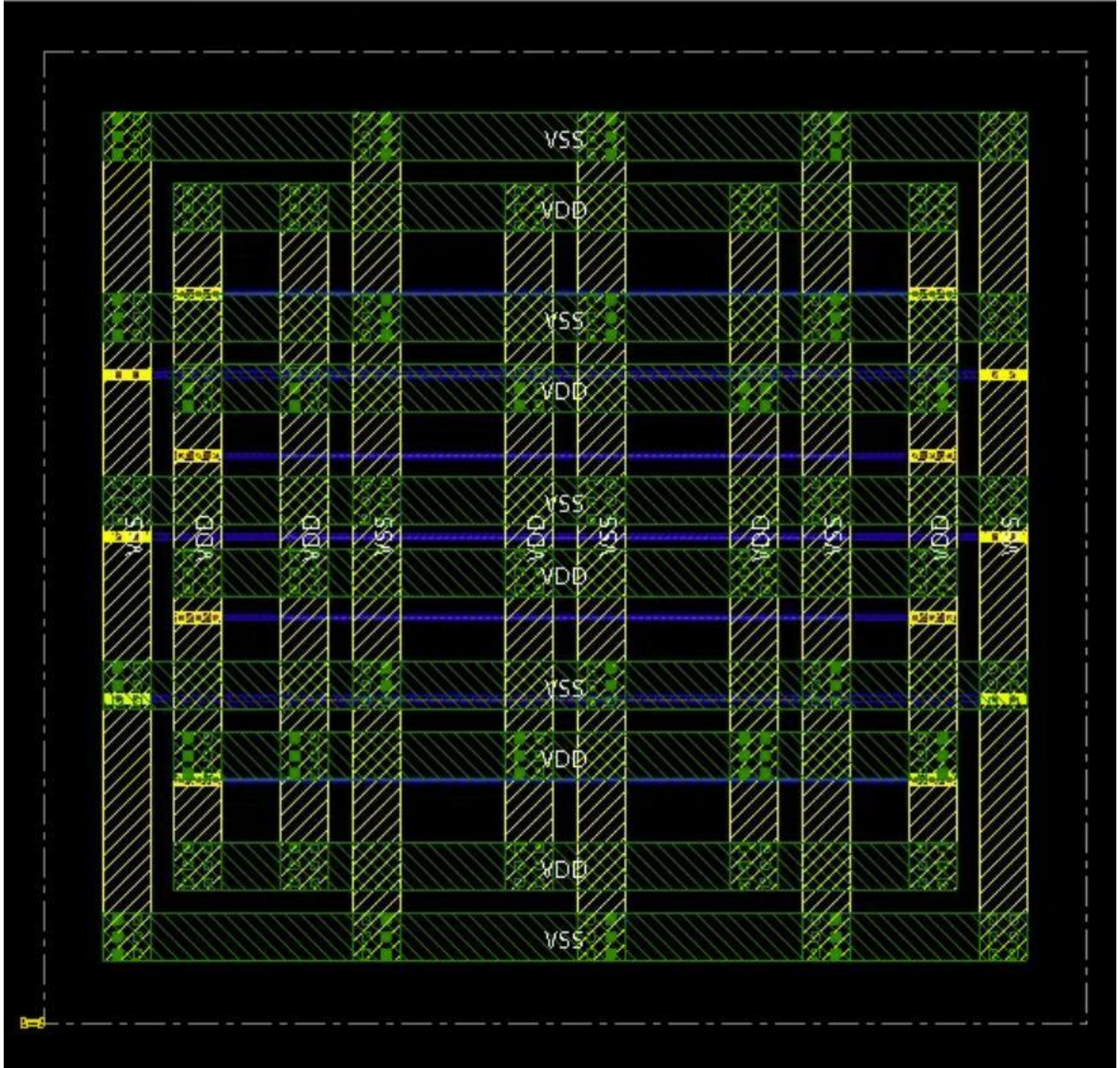
Horizontal Power Stripes



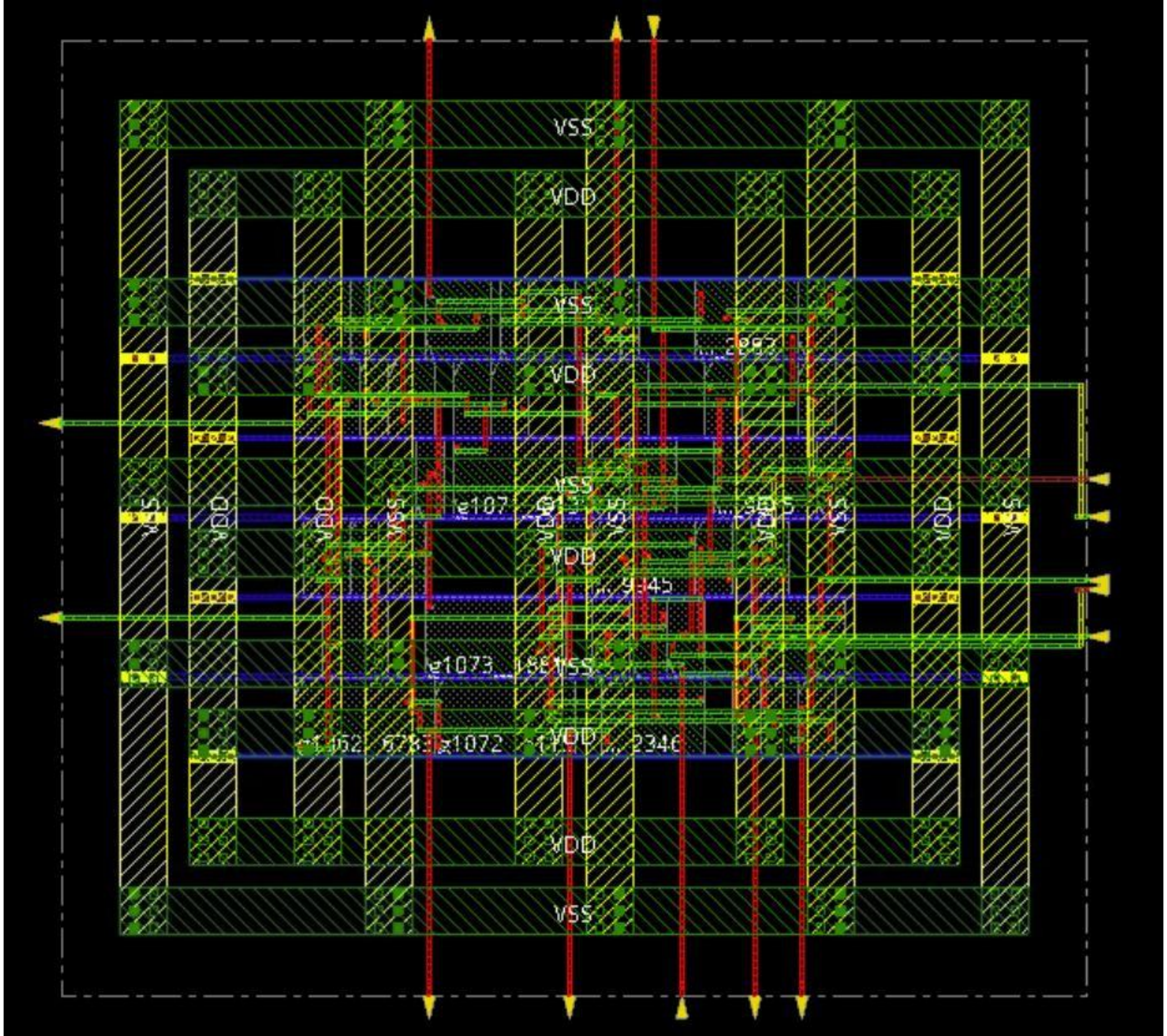
Vertical Power Stripes



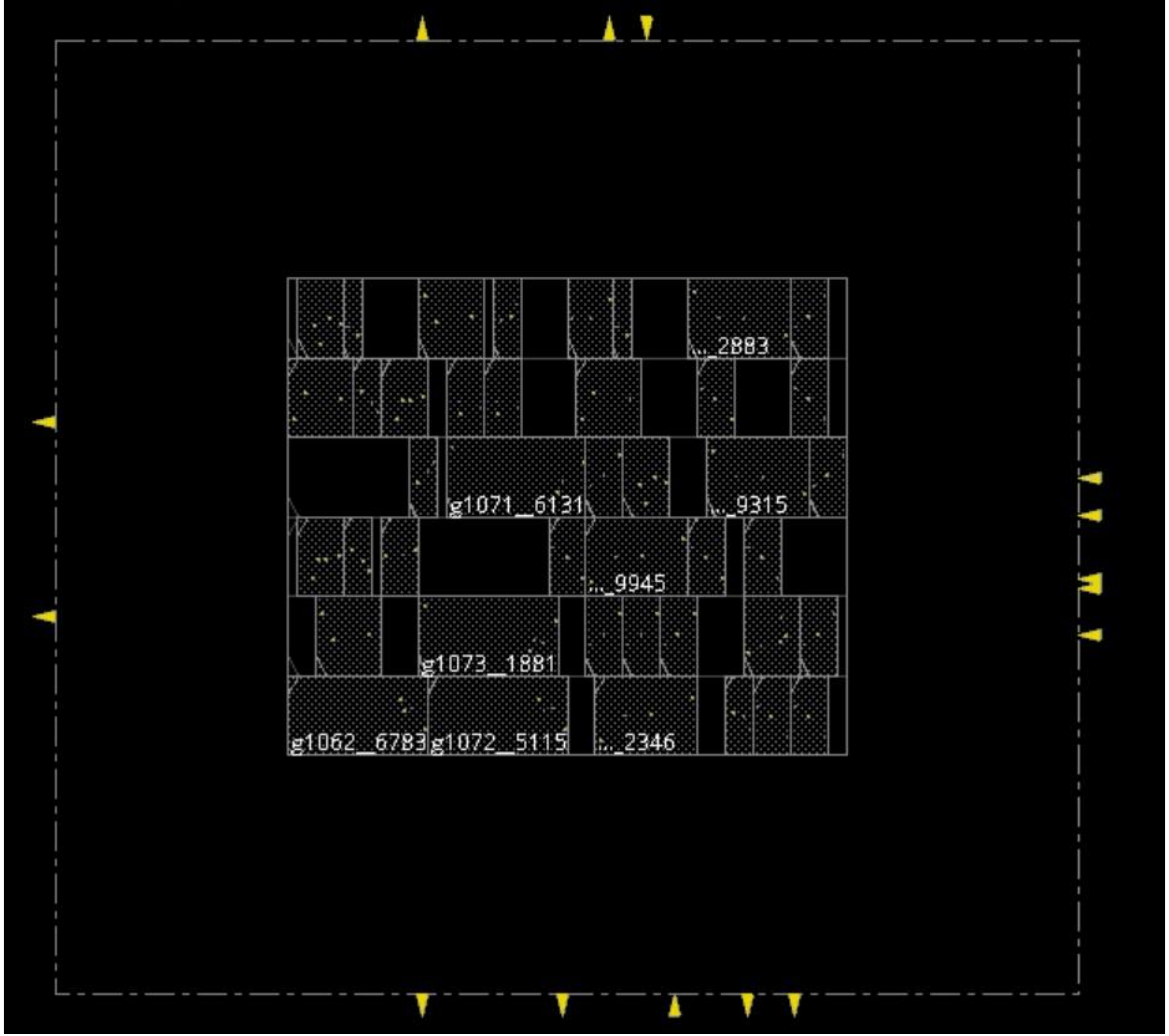
Routing



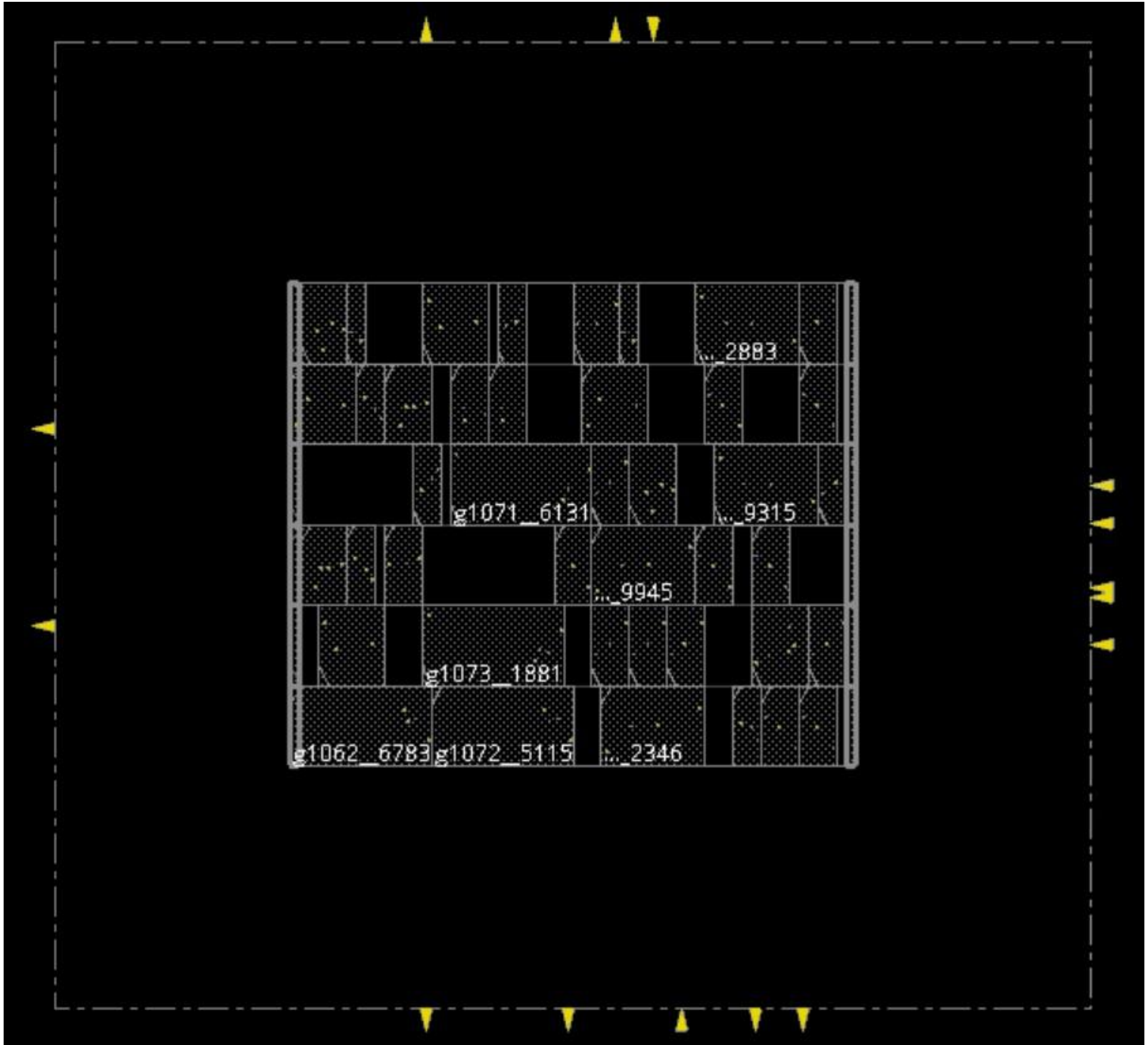
Placed Design :



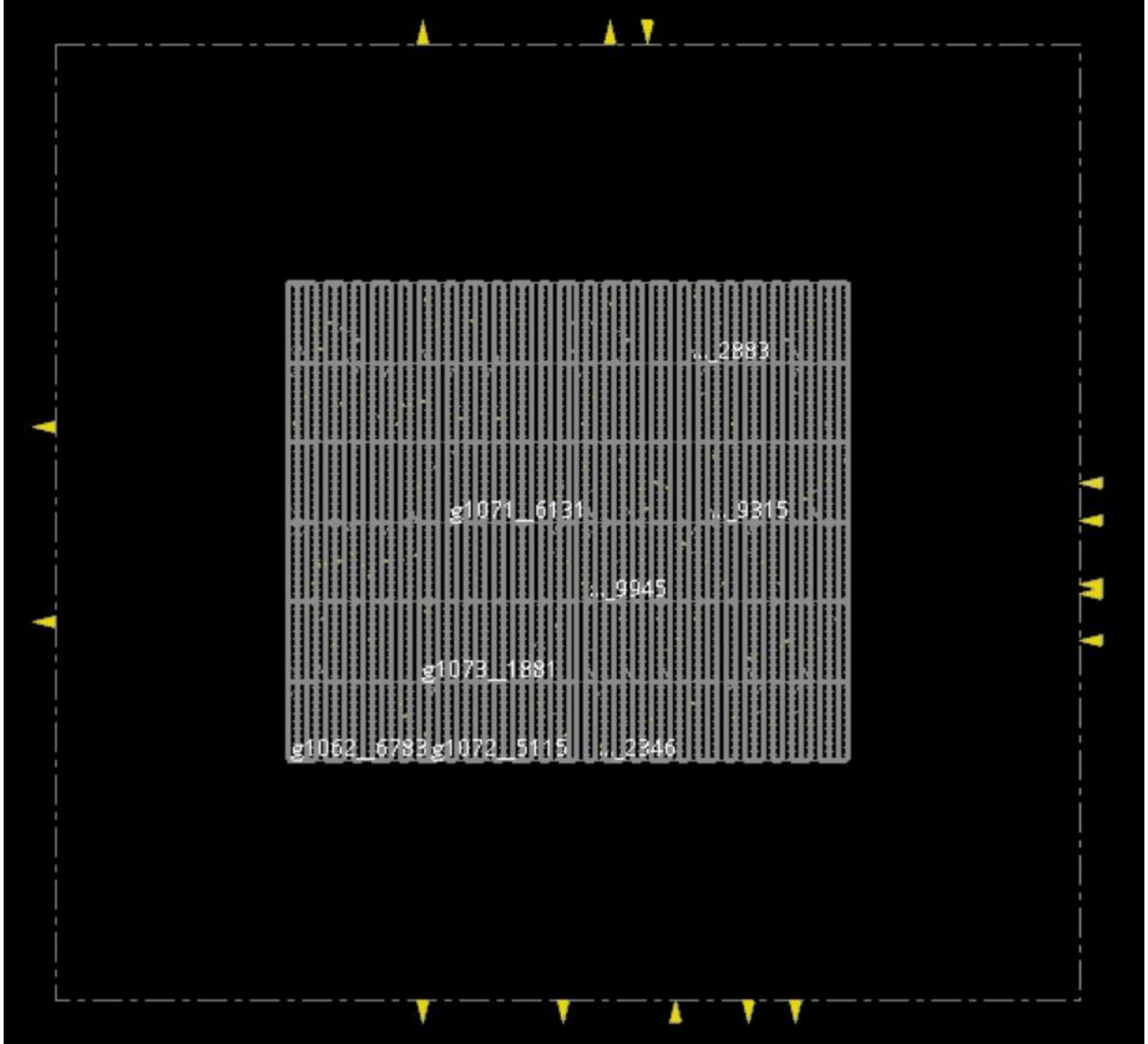
Standard Cell Placement



Standard Cell Placement with End cap :

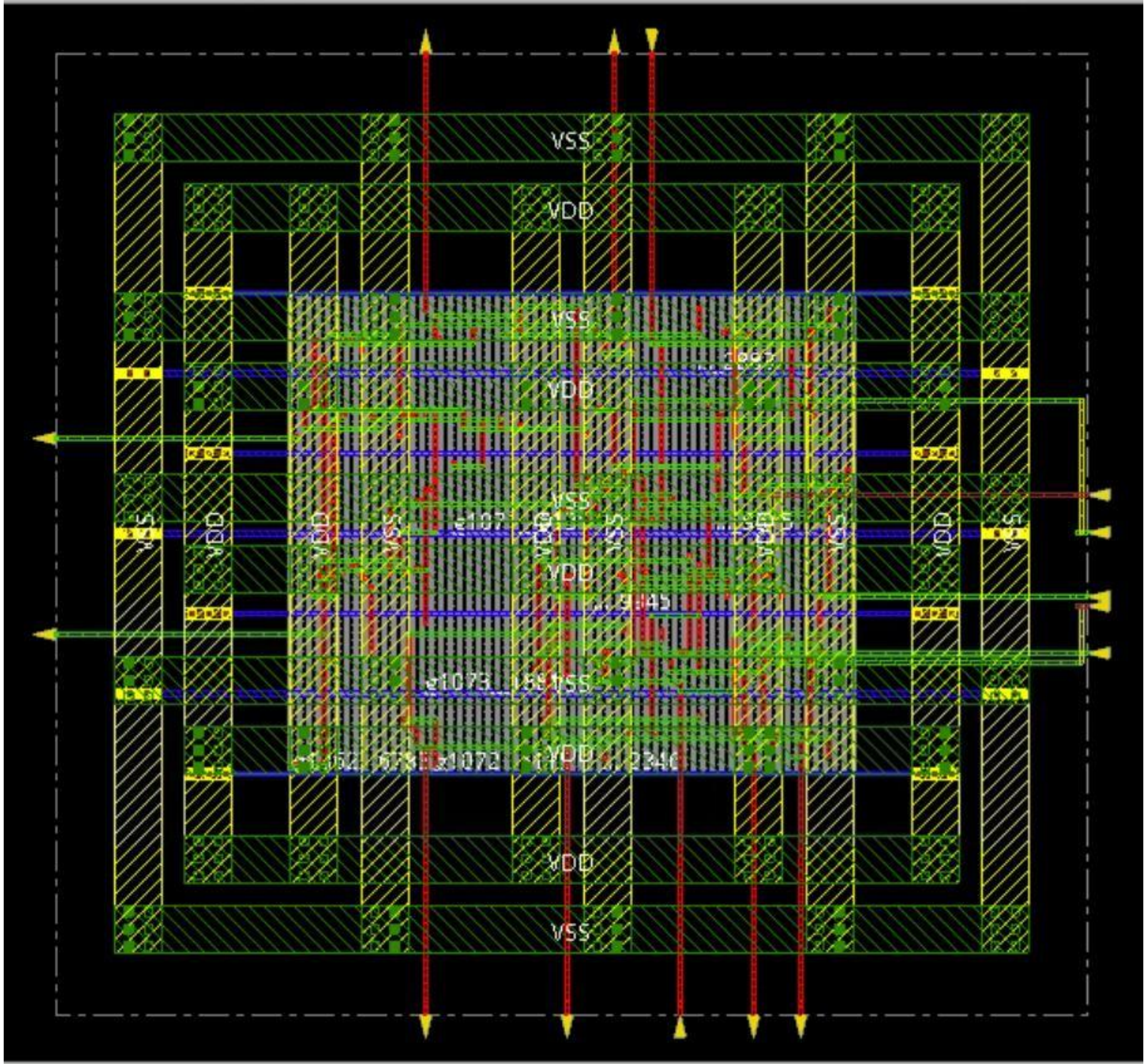


Well Tap :



..._2883
g1071_6131
..._9315
..._9945
g1073_1881
g1062_6783
g1072_5115
..._2346

Design with well Tap



Pre –routing Timing Summary

```
------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Setup views included:
 Worst_Case

+--------------------+---------+---------+
|     Setup mode     |   all   | default |
+--------------------+---------+---------+
|           WNS (ns):| 0.000   |  0.000  |
|           TNS (ns):| 0.000   |  0.000  |
|    Violating Paths:|    0    |    0    |
|          All Paths:|    0    |    0    |
+--------------------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 189.394%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------------
Reported timing to dir timingReports
Total CPU time: 0.12 sec
Total Real time: 1.0 sec
Total Memory Usage: 3773.691406 Mbytes
*** timeDesign #2 [finish] () : cpu/real = 0:00:00.1/0:00:00.9 (0.1), totSession cpu/real = 0:02:29.4/0:23:25.8 (0.1), mem = 3773.7M
```

Best_case

```
------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Hold views included:
 Best_Case

+--------------------+---------+---------+
|     Hold mode      |   all   | default |
+--------------------+---------+---------+
|           WNS (ns):|  0.000  |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |
|          All Paths:|    0    |    0    |
+--------------------+---------+---------+

Density: 189.394%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------------
Reported timing to dir timingReports
Total CPU time: 0.51 sec
Total Real time: 0.0 sec
Total Memory Usage: 3746.988281 Mbytes
*** timeDesign #4 [finish] () : cpu/real = 0:00:00.5/0:00:00.5 (1.0), totSession cpu/real = 0:02:57.4/0:27:53.6 (0.1), mem = 3747.0M
```

Optimized Timing Summary :

```
------------------------------------------------------------------
             Initial Summary
------------------------------------------------------------------

Setup views included:
 Worst_Case

+--------------------+---------+
|     Setup mode     |   all   |
+--------------------+---------+
|           WNS (ns):|   N/A   |
|           TNS (ns):|   N/A   |
|    Violating Paths:|   N/A   |
|          All Paths:|   N/A   |
+--------------------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 189.394%
------------------------------------------------------------------
**optDesign ... cpu = 0:00:03, real = 0:00:17, mem = 3014.6M, totSessionCpu=0:04:14 **
*** InitOpt #1 [finish] (place_opt_design #3) : cpu/real = 0:00:03.5/0:00:16.5 (0.2), totSession cpu/real = 0:04:14.0/0:36:09.1 (0.1), mem = 3918.7M
** INFO : this run is activating medium effort placeOptDesign flow
**Info: (IMPSP-307): Design contains fractional 3 cells.
**Info: (IMPSP-307): Design contains fractional 3 cells.
*** ExcludedClockNetOpt #1 [begin] (place_opt_design #3) : totSession cpu/real = 0:04:14.0/0:36:09.2 (0.1), mem = 3918.7M
*** Starting optimizing excluded clock nets MEM= 3918.7M) ***
*info: No excluded clock nets to be optimized.
```

Optimized Timing Summary for Setup and Hold

Both Negative slack and Violating Paths are zero

```
     optDesign Final Summary
------------------------------------------------------------------

Setup views included:
 Worst_Case
Hold views included:
 Best_Case

+--------------------+---------+---------+
|     Setup mode     |   all   | default |
+--------------------+---------+---------+
|           WNS (ns):|  0.000  |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |
|          All Paths:|    0    |    0    |
+--------------------+---------+---------+

+--------------------+---------+---------+
|     Hold mode      |   all   | default |
+--------------------+---------+---------+
|           WNS (ns):|  0.000  |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |
|          All Paths:|    0    |    0    |
+--------------------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 189.394%
```